Paper 2 (Mark Scheme)

1 (a) product of force and perpendicular distance from line of action of force to axis / pivot B1

(b) (i) (use principle of moments by taking moment about A,)
either a.c.w. moment = $T \times AD \sin 47^\circ$ or $T \times AD \cos 43^\circ$
(where $AD = 3.2 \sin (180^\circ - 53^\circ - 47^\circ) / \sin 47^\circ$ from sine rule)
or a.c.w. moment = $T \times 3.2 \cos 10^\circ$ or $T \times 3.2 \sin 80^\circ$ C1

c.w. moment = $11 \times 9.81 \times 6.5 \sin 53^\circ + 9.0 \times 9.81 \times (6.5 / 2) \sin 53^\circ$
(where $\sin 53^\circ$ can be replaced by $\cos 37^\circ$) C1

so $T = 250$ N A0

(ii) [Method 1 – component of force]
vertical: $F_y = (11.0 + 9.0) \times 9.81 - 250 \sin (180^\circ - 53^\circ - 47^\circ) = 152.8$ N
horizontal: $F_x = 250 \sin 80^\circ = 246.2$ N C1

$F_A = (F_x^2 + F_y^2)^{1/2} = 290$ N A1

[Method 2 – cosine rule]
$F_A^2 = 250^2 + (20 \times 9.81)^2 - 2 \times 250 \times (20 \times 9.81) \cos 80^\circ$ (M1)

$F_A = 290$ N (A1)

(iii) moment of the weight of light about A decreases, so tension in cable decreases M1

(since tension remains in the direction of CD,)
either use cosine rule to show the force at A is smaller when tension decreases
or show with a well-labelled diagram that force A is smaller A1

